

Data-Driven Cardiovascular-Kidney-Metabolic Phenotyping in HCHS/SOL via a Survey-Weighted Dirichlet Process Mixture Model

1 Rationale

Hispanic/Latino adults are the largest and fastest-growing minority population in the United States, and cardiovascular disease is their leading cause of death; yet, studied in aggregate, this population appears to have better cardiovascular outcomes than non-Hispanic individuals, a discrepancy that reflects not resilience but the loss of information inherent in treating a heterogeneous, multi-origin population as a single "Hispanic" risk category¹. The clearest illustration of that loss is the so-called Hispanic paradox: Hispanic adults have historically shown lower age-adjusted cardiovascular mortality than non-Hispanic White adults despite a substantially worse burden of the risk factors that are supposed to drive that mortality^{2;3}. That paradox dissolves as soon as the aggregate category is disaggregated: cause-specific mortality trends differ sharply by Hispanic/Latino background, with Puerto Ricans showing mortality patterns similar to non-Hispanic Whites while Mexicans show lower rates⁴; foreign-born Cubans, Mexicans, and Puerto Ricans have markedly higher cardiovascular mortality than their US-born counterparts⁵; and between 1999 and 2018, stroke mortality rose in Hispanic adults even as it stalled in non-Hispanic White adults, with heart failure mortality accelerating fastest in Hispanic adults under 65⁶. The kidney and metabolic side of the burden is more straightforwardly severe and shows no comparable paradox: within HCHS/SOL itself, chronic kidney disease affects 13.7% of participants, with more than a two-fold range across Hispanic/Latino background groups (7.4% in South Americans to 16.6% in Puerto Ricans)⁷, Hispanic ethnicity is independently associated with a substantially elevated risk of progression to end-stage renal disease even after accounting for diabetes^{8;9}, diabetes prevalence in HCHS/SOL ranges from 10.2% to 18.3% depending on background¹⁰, incident diabetes accrues at 22.1 cases per 1,000 person-years¹¹, and metabolic syndrome prevalence ranges more than two-fold, from 27% in South Americans to 41% in Puerto Rican women¹². Aggregating this population, whether into a single "Hispanic" label or, as we argue below, into a single ordinal risk stage, is precisely what has repeatedly been shown to mask the clinically actionable heterogeneity within it¹³, and unsupervised, data-driven approaches that let subgroups emerge directly from clinical and social data, rather than from investigator-imposed categories, have already begun to reveal cardiometabolic risk profiles that cut across conventional race/ethnicity groupings in the broader US population¹⁴. It is against this backdrop, a numerically large, rapidly growing, and demonstrably heterogeneous high-risk population for whom aggregate categories have repeatedly obscured rather than clarified risk, that the adequacy of any single risk-staging tool applied to Hispanic/Latino adults deserves particular scrutiny.

The tool most likely to be applied to this population going forward is Cardiovascular-Kidney-

Metabolic (CKM) syndrome, introduced by the American Heart Association (AHA) in 2023 to formalize the pathophysiological interconnection among metabolic risk factors, chronic kidney disease (CKD), and cardiovascular disease (CVD)¹⁵. Its staging system assigns every individual to one of five ordinal stages, from Stage 0 (no risk factors) to Stage 4 (established clinical CVD). The founding presidential advisory itself frames staging somewhat broadly, as a structure meant to "promote prevention across the life course," explicitly distinct from the companion "prediction algorithms" (the PREVENT equations) that the AHA commissioned separately to furnish continuous cardiovascular risk estimates¹⁶. In practice, however, the staging system has been adopted and validated almost entirely as a risk-stratification device: it is described as designed to "reflect the pathophysiology, spectrum of risk, and opportunities for prevention"¹⁷, reviewed as a framework that "allows targeted risk stratification and evidence-based management to interrupt CKM trajectories"¹⁸, and empirically validated on exactly that basis: in a 14.3-year population-based cohort, CKM stage showed a stepwise, monotonic association with cardiovascular disease and all-cause mortality, leading the authors to conclude that their findings "support the relevance of CKM staging for cardiovascular risk stratification and prevention in the general population"¹⁹. So whatever the precise balance the AHA advisory struck between staging as a prevention-organizing structure and staging as a graded risk tool, both the field's working interpretation and the subsequent validation literature treat, use, and confirm CKM stage as a risk-stratification instrument, on the premise that stage should track increasing cardiovascular and renal risk and thereby guide stage-specific prevention and treatment. It is precisely this stepwise property between stages that makes the internal composition of any single stage consequential: if a stage is to do stratifying work, the risk within it should be reasonably homogeneous, and a growing body of commentary has argued that CKM staging instead still leaves substantial phenotype-level heterogeneity for stratification to resolve²⁰, or that its value lies more in organizing multidisciplinary care than in fine-grained risk discrimination²¹. Applied to HCHS/SOL, the population-level pattern that motivates this proposal reappears within the staging tool itself: 50.6% of the cohort was classified as Stage 2 at baseline²², closely replicating an earlier HCHS/SOL estimate of 53.4%²³. This is not unique to HCHS/SOL, nor is it a sampling artifact. The same clumping in Stage 2 appears in the Dallas Heart Study (46%)²⁴, in a nationwide Japanese cohort of 266,256 adults (65.0%)²⁵, and among postmenopausal women in NHANES (55.7%)²⁶, and a recent review concluded that Stage 2 "represents the most prevalent category, affecting nearly half of adults in Western cohorts"¹⁸, a pattern that persists, with variation in degree, across NHANES race/ethnicity strata²⁷. The reason is structural rather than incidental: because any CKD, including mild albuminuria with normal estimated glomerular filtration rate, or the presence of a single metabolic risk factor is sufficient to qualify for Stage 2, the criteria function as a broad catch-all²⁸, and under the original definition all individuals with type 2 diabetes are uniformly classified as Stage 2 regardless of how well- or poorly-controlled their disease is²⁹. A staging system whose purpose is risk stratification loses much of its discriminative value once half of a population, spanning a genuinely wide range of underlying cardiometabolic and renal risk, is compressed into one category, and this is exactly the kind of aggregation that has already been shown, in the broader Hispanic/Latino cardiovascular literature, to mask clinically meaningful subgroup differences rather than reveal them.

This limitation has been explicitly recognized, and several groups have proposed fixes, but each works within the pre-specified AHA staging rather than questioning whether an investigator-imposed ordinal partition is the right representation of cardiometabolic risk to begin with. Tian et al.³⁰ developed a sex-specific algorithm to manually subdivide Stage 2 into lower- and higher-risk subgroups (2a/2b), achieving meaningfully different 10-year cardiovascular mortality risk between the two. Gohda et al.²⁹ integrated KDIGO eGFR/albuminuria risk categories into Stage 2 for

individuals with type 2 diabetes and showed improved discrimination for CKD progression relative to the unmodified stage. Masoumi et al.³¹ proposed a continuous CKM severity score as a complement to categorical staging, explicitly to mitigate the limitations of categorical staging, and demonstrated a substantial mortality-risk gradient within Stage 2 alone. Two unsupervised approaches have begun to probe this heterogeneity empirically: Yang et al.³² applied k -means clustering with a fixed number of clusters to UK Biobank participants in CKM Stages 1–3 and recovered five phenogroups with materially different long-term risk of stroke, chronic kidney disease, and liver disease, while Zhou et al.³³ used untargeted plasma metabolomics and unsupervised clustering to identify metabolic subtypes cross-cutting AHA-defined stages. A parallel commentary literature has begun to question the added value of the CKM construct itself relative to existing risk tools, asking whether it represents "a new frontier or simple rebranding"³⁴ and flagging specific structural gaps such as the omission of hepatic biomarkers despite their established role upstream of insulin resistance³⁵. Every one of these corrective efforts, however, is supervised by the AHA staging system: each starts from the pre-defined stages, most often Stage 2 alone, and attempts to split or score risk within that boundary, rather than asking what structure the joint distribution of cardiovascular, kidney, and metabolic biomarkers would reveal if no ordinal staging rule were imposed at all. Where unsupervised methods have been used, the number of clusters has been fixed a priori and the cohorts studied, UK Biobank, single-center metabolomics samples, do not reflect the ethnic and cardiometabolic heterogeneity documented within the U.S. Hispanic/Latino population and summarized above. A prior latent class analysis in HCHS/SOL identified four subclasses of metabolic syndrome expression³⁶, demonstrating that this cohort’s biomarker data do contain recoverable substructure, but that analysis used a small, pre-specified set of canonical metabolic syndrome components rather than the fuller panel of cardiac, renal, and metabolic biomarkers now organized under the CKM framework, used a parametric latent class model that still required the number of classes to be fixed and compared across candidate values, and was not linked to, or evaluated against, the subsequently introduced AHA CKM staging.

This motivates the present proposal: to derive cardiovascular-kidney-metabolic phenotypes for the entire HCHS/SOL cohort ($n=16,415$) directly from the joint distribution of CKM-relevant biomarkers, with no ordinal staging structure imposed and no number of phenotypes fixed in advance, and only afterward to compare the resulting data-driven phenotype assignments against each participant’s AHA CKM stage. If Stage 2 truly represents a biologically coherent, homogeneous risk category, the data-driven phenotypes should recover that homogeneity. If, as the sub-staging and continuous-score literature above suggests, Stage 2 instead pools together individuals with materially different multi-organ risk profiles, an unsupervised approach applied to the whole cohort should reveal that structure directly, without requiring an analyst to first guess which covariates might split the stage, and should do so in the one large, richly phenotyped cohort constructed specifically to represent the diversity of the U.S. Hispanic/Latino population rather than a single background group or a non-Hispanic reference population.

Two classes of methods dominate the existing cardiometabolic phenotyping literature: hard-partition algorithms such as k -means, used by Yang et al.³², and finite mixture or latent class models with a pre-specified or model-selected number of classes, used throughout the metabolic syndrome latent class analysis literature, including within HCHS/SOL itself^{36;37}. Both require the number of clusters K to be fixed, or chosen from a discrete candidate set via information criteria, before the clustering step is run. A Dirichlet process mixture model (DPMM) relaxes this requirement: it places a nonparametric prior over the partition of participants into clusters, allowing the number of supported phenotypes to be estimated jointly with the phenotypes themselves, and it returns full posterior uncertainty over cluster assignment rather than a single hard partition³⁸. DPMMs

have an established track record for exactly this kind of task, unsupervised phenotype discovery from a multivariate clinical marker profile without a pre-specified number of subtypes, including symptom-based phenotyping in Parkinson’s disease³⁹, profile-regression-style clustering of clinical covariates⁴⁰, and, most directly relevant, clustering of diabetes multimorbidity from electronic health record data, where DPMM was shown to adapt its cluster count to underlying data structure and to outperform fixed- K alternatives on high-dimensional, imbalanced clinical marker sets⁴¹. A concurrent, independently developed data-driven phenogroup classification of cardiometabolic risk⁴² further indicates that expanding beyond the canonical metabolic syndrome components to a richer risk-factor panel reveals phenogroups with materially different cardiovascular hazard, reinforcing that the CKM biomarker panel is rich enough to support this kind of discovery.

HCHS/SOL was not collected as a simple random sample: it is a two-stage, stratified, clustered area-probability sample of Hispanic/Latino households across four U.S. field centers, with unequal probabilities of selection by design, requiring dedicated sampling-weighted methodology even for standard analyses conducted within the cohort^{43;44}. Lin et al.⁴⁵ demonstrate explicitly, in this same cohort, that ignoring the complex sampling design in model-based analyses of HCHS/SOL data can bias inference, motivating design-aware estimation even for genetic association models. A DPMM fit naively to the observed HCHS/SOL sample, ignoring the sampling weights, would let over- or under-sampled strata, by center, background group, and other design strata, distort both the estimated number of phenotypes and the estimated size and biomarker profile of each phenotype, the same design-induced bias that has been documented broadly for Bayesian nonparametric models fit to complex survey samples without weighting^{46;47}. We therefore adopt a pseudolikelihood formulation in which each participant’s contribution to the DPMM likelihood is exponentiated by their normalized survey weight, so that the fitted mixture targets the sampling-weighted Hispanic/Latino population represented by HCHS/SOL rather than merely the realized sample^{48;49}. Exponentiating the likelihood by weights alone, however, is insufficient for valid inference: the within-cluster dependence induced by a multistage design inflates the effective information contributed by each unit relative to what plain weighting assumes, which in turn produces under-coverage of pseudo-posterior credible regions if left uncorrected. We address this using the sandwich-type Godambe information correction of Williams and Savitsky⁵⁰, which applies a fast, one-time post-processing projection to Markov chain Monte Carlo draws from the pseudo-posterior, rescaling and reshaping the pseudo-posterior covariance matrix so that nominal coverage of cluster-parameter uncertainty is approximately restored. This correction is implemented in the `csSampling` framework⁵¹ and has since been extended to multilevel model structures⁵², making it directly applicable to a hierarchically clustered design such as HCHS/SOL’s. In combination, the survey-weighted pseudolikelihood and Godambe-adjusted pseudo-posterior allow us to recover phenotypes, phenotype prevalences, and posterior uncertainty on cluster membership that are simultaneously representative of the target Hispanic/Latino population rather than the realized, design-biased sample, and accompanied by correctly calibrated uncertainty, rather than the artificially narrow credible intervals that a naive weighted pseudo-posterior would otherwise produce.

Applying an unweighted, fixed- K clustering algorithm, or simply re-deriving sub-strata within the existing AHA Stage 2 definition, would leave open the question motivating this proposal: is the concentration of half of the HCHS/SOL cohort in a single CKM stage a real reflection of homogeneous risk in this population, or an artifact of an ordinal staging rule that was not designed with the joint multivariate structure of cardiovascular, kidney, and metabolic biomarkers, or with a heterogeneous, multi-origin, historically underrepresented population, in mind? By phenotyping the entire cohort from the biomarker data directly, with the number of phenotypes determined by the data rather than fixed by the analyst, and with survey design and uncertainty properly

accounted for, and only subsequently cross-tabulating the resulting phenotypes against AHA CKM stage, this proposal provides a direct, population-representative empirical test of that question. Given the documented within-Stage-2 risk heterogeneity already reported using stage-conditioned sub-classification and continuous scoring approaches^{30;31}, and the burden and heterogeneity in cardiovascular, kidney, and metabolic risk across Hispanic/Latino background groups documented above^{7;12;13}, we anticipate that a fully data-driven phenotyping of HCHS/SOL will recover clinically distinct subgroups currently pooled under Stage 2. Because this population already bears a disproportionate and heterogeneous cardiovascular-kidney-metabolic disease burden that aggregate categories have repeatedly been shown to mask, and because it remains underrepresented in the cohorts that originally informed the AHA staging framework, resolving whether that framework’s coarsest stage reflects real homogeneity or an artifact of its design has direct and immediate implications for precision cardiovascular-kidney-metabolic prevention in Hispanic/Latino adults specifically.

Optional analysis: incorporating metabolomics. Three batches of metabolomics data exist within HCHS/SOL, but, having been generated as part of an ancillary study rather than the core examination, they are available for only a subset of participants rather than the full cohort. This partial coverage should not be allowed to constrain the primary analysis: the survey-weighted DPMM described above is fit on the clinical cardiovascular, kidney, and metabolic biomarker panel that is available for essentially the entire cohort, precisely so that every participant receives a phenotype assignment regardless of ancillary-study enrollment. Metabolomics is instead treated as an optional, secondary layer, for two reasons beyond incompleteness. First, three separate measurement batches introduce technical variation that, left uncorrected, can itself masquerade as biological clustering structure; before any downstream use, batch-level correction is required, using quality-control-sample-based signal correction of the kind developed for large-scale, multi-batch metabolic profiling studies⁵³, a removal-of-unwanted-variation approach designed specifically for combining metabolomics data across batches and platforms⁵⁴, or a comparable benchmarked batch-correction method⁵⁵; batch should also be retained as a covariate and checked against the final phenotype assignments. Second, ancillary-study enrollment is itself a selection mechanism that need not be random with respect to CKM phenotype, so any downstream use of the metabolomics subsample should not simply assume it is missing completely at random.

Two complementary strategies allow metabolomics to be incorporated once these prerequisites are addressed, and both preserve the primary, full-cohort phenotyping rather than replacing it. The simpler strategy is post hoc validation: within the metabolomics-available subsample, cross-tabulate the phenotype assignments already obtained from the biomarker-only DPMM against metabolomic profiles to test whether any recovered phenotype, particularly one overlapping the coarse AHA Stage 2 category, is further separable on metabolomic grounds, mirroring the logic already used by the CKM sub-staging literature discussed above but applied to data-driven rather than investigator-defined phenotypes. The more ambitious strategy is to extend the joint model itself: rather than restricting analysis to complete cases or imputing metabolomics for the majority of the cohort that never had it measured, a further block can be added to the DPMM likelihood whose contribution is included only for participants with observed metabolomics and simply omitted, not imputed, for everyone else, following the general approach to Bayesian integrative modeling of multi-omics data with block-wise missingness^{56;57}. A related and more explicitly nonparametric alternative is to let the metabolomics subsample support its own clustering that adheres loosely to, rather than being forced to coincide with, the consensus phenotyping obtained from the full-cohort biomarker panel, in the spirit of Bayesian consensus clustering for multiple, unequally covered data sources⁵⁸ and

its recent extension to a fully nonparametric, two-level Dirichlet process construction that similarly avoids prespecifying the number of clusters in either source⁵⁹; partial multi-omics clustering methods developed for the analogous problem of cancer subtyping with incomplete omics coverage, which explicitly avoid imputing missing platforms, offer a further precedent⁶⁰. Naively imputing metabolomics values for the full cohort before clustering is deliberately avoided under either strategy, since imputation-based approaches to missing rows in multi-omics integration have been shown to produce rapidly growing uncertainty in the resulting structure as the proportion of missingness increases⁶¹, which is precisely the regime an ancillary substudy occupies. Finally, because ancillary-study membership constitutes a second, non-random sampling phase nested within the primary HCHS/SOL design, any analysis of the metabolomics subsample should reweight participants by the inverse probability of ancillary-study selection, following established two-phase survey methodology⁶² and approaches for borrowing information from auxiliary data available on only part of a cohort⁶³, combined multiplicatively with the primary HCHS/SOL design weight rather than substituted for it; comparable inverse-probability-weighting corrections have been shown to materially reduce selection bias in omics analyses restricted to non-randomly available subsamples⁶⁴.

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